

Endpoint Control Does Not Identify Temporal Causal Factorization

Abstract

Cognitive experiments often show that changing an action, cue, control signal, or policy changes a later outcome. Such endpoint effects establish causal capacity, but they need not identify which earlier time-indexed intervention distinctions remain causally relevant at that endpoint. We formalize this identification problem by treating a temporally factored intervention history as a compound input to an endpoint channel. Arbitrary relabeling of complete histories preserves endpoint-channel capacity, whereas questions about a fixed temporal coordinate require a narrower factor-preserving equivalence. The mismatch implies that endpoint control cannot, in general, identify temporal causal factorization. A three-step deterministic witness gives abstractly equivalent endpoint channels in which the first intervention coordinate has full support in one system and no endpoint support in the other. The distinction survives a shared stochastic post-channel while the relevant endpoint contrast remains detectable. Exhaustive enumeration of small deterministic systems further shows substantial structural ambiguity within matched endpoint-capacity classes. The result is not a new theory of cognition or a replacement for causal inference, credit assignment, or temporally resolved control measures. It is a bounded methodological result: endpoint-level causal capacity and temporal causal relevance answer different identification questions, a distinction that matters when cognitive theories interpret delayed outcomes, sequential decisions, or learning histories.

Keywords: causal abstraction; temporal credit assignment; identifiability; endpoint control; empowerment; sequential decision making; causal learning

1. Introduction

Cognitive experiments frequently infer causal or functional relevance from endpoints: an action changes a later choice, a cue changes a delayed judgment, a control signal changes eventual performance, or a learning manipulation changes a final response. These findings can establish that an intervention matters to an outcome while leaving a different question unresolved: which temporally located distinctions in the preceding process still constitute the causal difference observed at the endpoint?

That distinction is closely related to the credit-assignment problem. In reinforcement learning and decision research, credit assignment asks which earlier actions or states should be associated with later outcomes; human work further shows that task structure and causal relatedness can shape that attribution (Moran, Dayan, & Dolan, 2021). Contemporary reviews likewise cast temporal credit assignment as learning the influence of an action on an outcome across delayed and noisy histories (Pignatelli et al., 2024). The present paper does not propose a new credit-assignment algorithm. It asks what an endpoint-level control description can identify before any such cognitive mechanism is posited.

Information-theoretic formulations of control provide a useful test case because they deliberately compress possibilities. Empowerment quantifies potential causal influence as the capacity of an action-to-future-state or action-to-future-observation channel, and recent cognitive work has used empowerment-related controllability and variability as cues in human causal learning (Yiu, Allen, Ginosar, & Gopnik, 2026). Richer causal, temporal, and process-level formalisms can represent temporal organization. The question here is therefore not whether cognitive science lacks temporally resolved tools, but what is lost when a temporally factored intervention history is treated only as an atomic compound input to an endpoint channel.

The narrower question examined here is one of identification.

Suppose a sequence

$$A_0, A_1, \dots, A_{T-1}$$

is treated as a compound channel input

$$U = (A_0, \dots, A_{T-1}),$$

and its influence on a later endpoint state E_T is summarized by an endpoint-control quantity. What information about the temporal identity of the constituent interventions remains recoverable from that abstraction?

We show that the answer is incomplete in a precise sense. Endpoint channel capacity is invariant under arbitrary relabeling of compound input histories. The raw temporal-support object transforms equivariantly under the narrower class of relabelings that preserve temporal factorization, while its time-preserving orbit is invariant under those relabelings. The larger symmetry of the endpoint-control abstraction therefore identifies systems that need not be equivalent with respect to which earlier intervention coordinates remain causally consequential. For cognitive interpretation, this means that matching endpoint control does not by itself license matching temporal causal credit or provenance.

The result can be expressed compactly as

Endpoint control identifies causal capacity but not, in general, its temporal factorization.

This is not primarily a criticism of endpoint control. The information is absent because the abstraction is doing exactly what it was designed to do: ignoring distinctions among input labels that do not affect communication capacity. The same invariance becomes insufficient only when the scientific question concerns the temporal provenance of the causal distinctions composing that capacity. A cognitive experiment may therefore be perfectly informative about endpoint influence while remaining non-identifying for a temporally indexed causal question.

We establish the point constructively, generalize it to stochastic channels, quantify its finite combinatorial extent, and locate the result relative to causal abstraction, empowerment, temporal credit assignment, and related cognitive-science formalisms. The formal result is architecture-neutral; the cognitive-science payoff is a constraint on what endpoint observations can support as an inference about temporally distributed processes.

2. Methodological scope and system boundary

Consider a system H coupled to an environment E , with states

$$H_t, E_t.$$

The analysis concerns physical or causal system-environment dynamics and is deliberately architecture-neutral. It does not require a conscious narrator, an internal representation of future possibilities, or a dedicated decision module. If conscious content exists, it is a separate observation channel and is not identified with system state:

$$H_t \neq C_t.$$

Nothing in the formal development requires C_t , and it is omitted below.

Let

$$A_t \in \mathcal{A}_t$$

index distinguishable system-side interventions or perturbations at time t . The notation A_t should not be read as requiring conscious action selection. It denotes an analyst-defined intervention coordinate capable of altering the system-environment evolution.

For a fixed initial condition, the resulting endpoint state is represented by an intervention kernel

$$K_T(e \mid do(a_0, \dots, a_{T-1})).$$

In the deterministic case this reduces to a function

$$F_T: A_0 \times \dots \times A_{T-1} \rightarrow E_T.$$

The system boundary matters because the causal interpretation of an intervention is boundary-relative. The present analysis takes that boundary as declared and asks what can be inferred after a temporally labeled intervention sequence has been compressed into an endpoint channel. Cognitive interpretation is warranted only when the factors correspond to scientifically meaningful temporal intervention coordinates - for example, ordered actions, cues, control signals, or experimentally defined perturbations.

The analysis does not address the separate causal-identification question of whether an observed trajectory requires intact cross-boundary interaction. Identical observational trajectories may arise from internally generated versus coupling-dependent dynamics, and distinguishing those cases requires interventions on the relevant causal pathways. That general distinction is already well established in intervention-based causal analysis.

2.1 Generative AI use

OpenAI ChatGPT was used during the research and manuscript-development workflow for research assistance, including conceptual formalization, literature discovery and organization, adversarial critique and stress testing, drafting and editing, notation refinement, simulated peer-review workflows, and submission preparation. Use occurred across multiple sessions, including submission preparation in September 2026; no single model version is represented as having been used throughout the project. Human authorship originated and directed the research questions, adjudicated outputs, retained intellectual control and final scientific judgment, and reviewed and verified cited sources and factual claims used in the manuscript. Generative AI is not an author and generated no participant data or empirical dataset. No sensitive participant data were provided to the tool.

3. Endpoint control, factorization, and target structure

For a multidisciplinary reader, the formal argument separates three objects that can otherwise be conflated: a time-labeled intervention history, an aggregate endpoint-control quantity, and a time-indexed causal-support target. The proof asks whether the first can be compressed into the second without losing information needed to identify the third.

3.1 Time-factored intervention histories

Let the finite intervention alphabet at time coordinate t be A_t . The complete intervention-history space is the Cartesian product

$$U_T = \prod_{t=0}^{T-1} A_t. \tag{2}$$

A realized history is written $u = (a_0, \dots, a_{T-1})$, with random-variable notation $U = (A_0, \dots, A_{T-1})$. The factorization in Equation (2) is part of the declared scientific description: the coordinates are not merely positions in an arbitrary label string but denote interventions associated with particular temporal indices.

For a fixed initial condition, the fully time-labeled endpoint intervention kernel is

$$K_T(e \mid do(a_0, \dots, a_{T-1})). \tag{3}$$

Equivalently, after the complete history u is treated as one compound input symbol, the same numerical kernel can be written as the ordinary channel

$$K_T: U_T \rightarrow \Delta(E_T), u \mapsto K_T(\cdot \mid do(u)). \quad (4)$$

The distinction between Equations (3) and (4) is representational rather than dynamical. Equation (3) retains the meanings of the temporal coordinates; Equation (4), if considered only as an abstract channel over atomic input symbols, need not retain that embedding.

3.2 Endpoint channel capacity

The endpoint channel capacity is

$$C_T(K) = \max_{p(u)} I(U; E_T). \quad (5)$$

It measures the maximum distinguishable influence that can be transmitted from the compound intervention history to the endpoint. If $E_T = F_T(U)$ deterministically for a map $F_T: U_T \rightarrow E_T$, then

$$C_T(F_T) = \log_2 |\text{Im}(F_T)|. \quad (6)$$

Thus deterministic endpoint capacity depends on the number of reachable endpoint classes, not on how temporally factored histories are arranged inside those classes.

3.3 Time-indexed causal support

Fix a coordinate i , a setting u_{-i} of all other coordinates, and two distinct values $a_i, a_i' \in A_i$. A quantitative controlled endpoint contrast may be written

$$\delta_{i,T}(u_{-i}; a_i, a_i') = D[K_T(\cdot \mid do(u_{-i}, a_i)), K_T(\cdot \mid do(u_{-i}, a_i'))], \quad (7)$$

where D is any application-appropriate divergence or statistical distance; no particular choice of D is introduced here. The qualitative coordinate-labeled intervention-effect support relation is

$$S_{i,T}(K) = \{(u_{-i}, a_i, a_i') : K_T(\cdot \mid do(u_{-i}, a_i)) \neq K_T(\cdot \mid do(u_{-i}, a_i'))\}. \quad (8)$$

Here "support" means support of a controlled causal/intervention effect: it is not the support of a probability distribution.

For deterministic F_T , membership in $S_{i,T}$ is equivalent to

$$F_T(u_{-i}, a_i) \neq F_T(u_{-i}, a_i'). \quad (9)$$

Equality means that the two histories have reconverged with respect to the endpoint under that controlled contrast.

Marginal mutual information $I(A_i; E_T)$ is not equivalent to Equation (8). In the standard XOR synergy example, if $E_T = A_i \oplus A_j$ under independent uniform binary interventions, then $I(A_i; E_T) = 0$ although changing A_i while holding A_j fixed always changes the endpoint. This establishes only that marginal mutual information need not coincide with the controlled support relation; it is not a claim that every conditional or multivariate information formulation fails.

The complete stochastic object remains the time-labeled intervention kernel in Equation (3). The support relation in Equation (8) records only the qualitative controlled distinctions needed for the present identification question. Because the labels inside Equation (8) themselves may be relabeled, the raw set is not treated as an invariant object under time-preserving relabeling. Section 4 defines the induced action and the corresponding structural target.

4. Two equivalence structures

The identification result follows from comparing two transformation groups acting on the same compound history space.

4.1 Abstract compound-input equivalence

Let $\sigma: U_T \rightarrow U_T$ be any bijection and define the row-permuted channel

$$K^\sigma(e | u) = K(e | \sigma(u)). \quad (10)$$

All such relabelings form

$$G_{abstract} = \text{Sym}(U_T). \quad (11)$$

Two endpoint channels are abstract-input equivalent, written $K \sim_{abstract} K'$, when $K' = K^\sigma$ for some $\sigma \in G_{abstract}$. Channel capacity is constant on each resulting equivalence class because the optimizing input distribution can be permuted with the channel rows. Any other endpoint-channel statistic that is invariant to compound-input labels has the same property.

4.2 Time-preserving equivalence and induced support action

Assume throughout that each finite factor A_i is nonempty. The declared factorization in Equation (2) admits a narrower class of relabelings. If temporal coordinates have fixed meanings, an intervention-value relabeling may act independently inside each factor:

$$g(a_0, \dots, a_{T-1}) = (g_0(a_0), \dots, g_{T-1}(a_{T-1})), g_i \in \text{Sym}(A_i). \quad (12)$$

These transformations form

$$G_{time} = \prod_{i=0}^{T-1} \text{Sym}(A_i). \quad (13)$$

The inclusion $G_{time} \subseteq G_{abstract}$ is always valid. It is strict whenever at least two temporal factors have cardinality greater than one. In the three-coordinate binary constructions below, this nondegeneracy condition is automatically satisfied. If only one factor is non-singleton, the product action can coincide with the full permutation group on U_T , so strictness must not be inferred from $T > 1$ alone.

We write $K \sim_{time} K'$ when $K' = K^g$ for some $g \in G_{time}$. The same group induces a relabeling action on each raw support set. For

$$s = (u_{-i}, a_i, a_i') \in S_{i,T}(K),$$

define

$$g \cdot s = \left((g_j(a_j))_{j \neq i}, g_i(a_i), g_i(a_i') \right). \quad (14)$$

With the channel convention $K^g(e \mid u) = K(e \mid g(u))$, the support sets transform equivariantly as $S_{i,T}(K^g) = g^{-1} \cdot S_{i,T}(K)$. Accordingly, the factor-sensitive target used below is not literal equality of the labeled sets but the G_{time} -orbit of the support family

$$S_T(K) = (S_{0,T}(K), \dots, S_{T-1,T}(K)), [S_T(K)]_{G_{time}}. \quad (15)$$

This orbit retains structural properties that cannot be altered by merely renaming intervention values inside fixed temporal coordinates. In particular, emptiness, fullness, and cardinality of a coordinate support set are preserved under the induced action.

The quotient of channels by $\sim_{abstract}$ therefore identifies at least as many concrete time-labeled systems as the quotient by $\sim_{time}$, and generally more under the stated nondegeneracy condition. An arbitrary history permutation may exchange or mix distinctions that belong to different temporal coordinates. This is the precise location of the potential information loss: not in the full kernel, but in passing to an equivalence class that forgets how compound input symbols are embedded in the temporal product structure.

5. Factorization-sensitive non-identifiability

The logical schema used here is standard group-invariance/orbit reasoning: an invariant statistic is constant on group orbits, so if a scientific target varies within an orbit induced by an abstraction, no statistic that is invariant on that orbit can recover the target. Maximal-invariant theory formalizes this relationship between group actions, orbits, and the information retained by invariant representations (e.g., Chiu, Sharp, & Bloem-Reddy, 2025). Proposition 1 is therefore an endpoint-control specialization of established quotient/orbit logic, not a claim to a new generic invariant-theory theorem.

Proposition 1 (endpoint-control invariance mismatch). Let Φ be an endpoint-channel statistic defined on channels over U_T and invariant under abstract compound-input relabeling:

$$\Phi(K^\sigma) = \Phi(K) \text{ for every } \sigma \in G_{abstract}. \quad (16)$$

Let the factor-sensitive target be the time-preserving orbit $[S_T(K)]_{G_{time}}$ defined in Equation (15). If there exist a channel K and a permutation $\sigma \in G_{abstract}$ such that

$$\left[S_T(K^\sigma) \right]_{G_{time}} \neq \left[S_T(K) \right]_{G_{time}}, \quad (17)$$

then Φ does not identify the factor-sensitive support target in general.

Proof. By Equation (16), K and K^σ have the same value of Φ , because they lie in the same abstract-input orbit. By Equation (17), the same two channels correspond to different time-preserving orbits of the support family. Therefore at least two channels that are indistinguishable to Φ have different values of the factor-sensitive target. No function of $\Phi(K)$ alone can reconstruct that target for all channels in the stated domain. $\square$

The proposition is deliberately conditional on the existence of the stated pair. Section 6 supplies the explicit witness. Its generic orbit logic is established; the paper's scientific claim is that the endpoint-control abstraction and the temporal-factorization target instantiate that logic in the stated way.

Corollary 1. Endpoint channel capacity is non-identifying for the factor-sensitive support target in general.

Proof. Capacity satisfies Equation (16) under arbitrary input-row permutation, and the construction in Section 6 satisfies Equation (17). $\square$

Corollary 2. More generally, any representation that identifies endpoint channels solely up to arbitrary compound-input relabeling cannot, without additional factorization information, recover every structural property of the intervention-effect support family that differs between time-preserving support orbits inside one abstract-input orbit.

This is stronger than the generic observation that a scalar statistic loses information. The failure can remain even when the endpoint channel is retained up to abstract input-symbol identity: the missing datum is the embedding of those atomic symbols into $A_0 \times \dots \times A_{T-1}$.

Conversely, nothing in Proposition 1 says that temporal information is absent from the full causal model. If the complete kernel $K_T(e \mid do(a_0, \dots, a_{T-1}))$ and the meanings of its coordinates are retained, every qualitative contrast in Equation (8) can be read directly from the model. The loss is induced by the coarser abstraction or quotient.

6. Explicit three-step deterministic construction

Consider three independent binary intervention coordinates

$$A_0, A_1, A_2 \in \{0, 1\}.$$

Define the first two environmental states identically for two systems:

$$E_1 = A_0$$

and

$$E_2 = (A_0, A_1).$$

At the final step define system P by

$$E_3^P = (A_0, A_2),$$

and system M by

$$E_3^M = (A_1, A_2).$$

The intervention histories are

$$U_3 = \{0, 1\}^3.$$

Shorter witnesses can establish the bare orbit mismatch; this three-step construction is retained because it makes persistence versus reconvergence explicit while also permitting the auxiliary trajectory-level checks below. In cognitive-science terms, the coordinates can be read as experimentally defined events in a sequential task; no claim is made that the binary construction is itself a psychological process model.

A complete endpoint table makes the construction directly checkable. Endpoint pairs are shown in the coordinate order used by each system.

a_0	a_1	a_2	$E_3^P = (a_0, a_2)$	$E_3^M = (a_1, a_2)$
0	0	0	(0,0)	(0,0)
0	0	1	(0,1)	(0,1)
0	1	0	(0,0)	(1,0)
0	1	1	(0,1)	(1,1)
1	0	0	(1,0)	(0,0)
1	0	1	(1,1)	(0,1)
1	1	0	(1,0)	(1,0)
1	1	1	(1,1)	(1,1)

6.1 Matched endpoint quantities

Both systems possess

$$|E_1| = 2, |E_2| = 4, |E_3| = 4.$$

Therefore their deterministic horizon-wise endpoint capacities are identical:

$$(C_1, C_2, C_3) = (1, 2, 2) \text{ bits}.$$

Under independent uniform probing,

$$(H(E_1), H(E_2), H(E_3)) = (1, 2, 2) \text{ bits}$$

for both systems.

Their full trajectories also have equal entropy:

$$H(E_1, E_2, E_3) = 3 \text{ bits}.$$

Moreover, the complete environmental trajectory identifies the complete three-bit intervention sequence in either construction: A_0 is read from E_1 , A_1 from the second component of E_2 , and A_2 from the final endpoint.

Therefore

$$I(A_{0:2}; E_{1:3}) = 3 \text{ bits}.$$

Thus both aggregate endpoint and aggregate trajectory quantities can be matched. These trajectory-level matches are auxiliary robustness checks against nearby aggregate summaries; Proposition 1 does not require them.

6.2 Different temporal support

In P ,

$$F_P(0, a_1, a_2) \neq F_P(1, a_1, a_2)$$

for every a_1, a_2 .

Therefore the first intervention coordinate remains endpoint-consequential:

$$S_{0,3}(P) \neq \emptyset.$$

Indeed every controlled A_0 contrast changes the endpoint.

In M ,

$$F_M(0, a_1, a_2) = F_M(1, a_1, a_2)$$

for every a_1, a_2 .

Therefore

$$S_{0,3}(M) = \emptyset.$$

The A_0 distinction is present earlier in the environmental trajectory but has fully reconverged with respect to E_3 .

Under the uniform probe this particular construction also gives

$$I(A_0; E_3^P) = 1$$

and

$$I(A_0; E_3^M) = 0.$$

Those mutual-information values illustrate the distinction but do not define it.

6.3 The stronger equivalence

Define

$$\sigma(a_0, a_1, a_2) = (a_1, a_0, a_2).$$

Then

$$F_M(u) = F_P(\sigma(u)).$$

System M is therefore obtained from P by permuting the rows of the compound endpoint channel.

Viewed as abstract communication channels,

$$P \cong M.$$

Viewed as time-factored causal systems,

$$P \approx_{time} M.$$

The permutation exchanges the temporal identities of A_0 and A_1 and therefore lies outside the time-preserving relabeling group. Under the repaired structural target, the decisive distinction remains: $S_{0,3}(P)$ is full whereas $S_{0,3}(M) = \emptyset$. No within-coordinate value relabeling can transform a full support set into an empty one, so the two support families occupy different G_{time} -orbits.

The three-step construction consequently demonstrates more than equal capacity:

Abstract endpoint-channel equivalence does not imply temporal-support equivalence.

7. Stochastic extension

The result does not depend on deterministic dynamics.

Let

$$Q(y \mid z)$$

be any stochastic channel placed after the deterministic endpoint variable z .

Define

$$K_P(y \mid u) = Q(y \mid F_P(u))$$

and

$$K_M(y \mid u) = Q(y \mid F_M(u)).$$

Since

$$F_M(u) = F_P(\sigma(u)),$$

we obtain

$$K_M(y \mid u) = K_P(y \mid \sigma(u)).$$

Therefore K_M remains a row permutation of K_P , implying

$$C(K_M) = C(K_P).$$

Indeed every abstract channel statistic invariant under input-symbol relabeling remains equal.

Yet temporal intervention support can still differ.

If

$$Q(\cdot \mid (0, a_2)) \neq Q(\cdot \mid (1, a_2)),$$

then in P ,

$$K_P(\cdot \mid 0, a_1, a_2) \neq K_P(\cdot \mid 1, a_1, a_2),$$

whereas in M ,

$$K_M(\cdot \mid 0, a_1, a_2) = K_M(\cdot \mid 1, a_1, a_2).$$

Thus the first-coordinate distinction remains causally detectable through the noisy channel in P but remains absent from M .

7.1 Symmetric-noise example

Let the four deterministic endpoint states pass through a four-symbol symmetric channel:

$$Q_\epsilon(y | z) = \begin{cases} 1 - \epsilon, & y = z, \\ \epsilon/3, & y \neq z. \end{cases}$$

For $0 \leq \epsilon \leq 3/4$, the degrading-to-uniform regime of this four-state symmetric channel, its capacity is

$$C_\epsilon = 2 - h_2(\epsilon) - \epsilon \log_2 3,$$

where h_2 is binary entropy.

Both P and M have exactly this same endpoint capacity.

Using total variation to quantify the controlled A_0 contrast gives

$$\delta_{0,3}^P = 1 - \frac{4\epsilon}{3},$$

whereas

$$\delta_{0,3}^M = 0.$$

At

$$\epsilon = \frac{3}{4},$$

all output rows become identical and the endpoint channel carries no information. The causal contrast in P then appropriately disappears.

Hence the factorization result persists throughout the informative noisy regime.

This separates the result from any peculiarity of deterministic reachable-state counting.

8. Exhaustive finite enumeration

The explicit three-step pair proves existence but does not indicate how much structure endpoint capacity may collapse.

We therefore exhaustively enumerate deterministic mappings from three binary intervention coordinates to four equally occupied endpoint classes.

There are

$$2^3 = 8$$

intervention histories.

A balanced mapping assigns exactly two histories to each of four labeled endpoint states. The number of such mappings is

$$\frac{8!}{(2!)^4} = 2520.$$

Under independent uniform probing, every one has

$$H(E_3) = 2 \text{ bits}$$

and deterministic endpoint capacity

$$C_3 = 2 \text{ bits}.$$

8.1 Marginal dependence

Across the 2,520 balanced labeled mappings,

$$I(A_0; E_3) = 0$$

for 576 mappings,

$$I(A_0; E_3) = 0.5$$

for 1,728 mappings, and

$$I(A_0; E_3) = 1$$

for 216 mappings.

Thus identical endpoint entropy and capacity coexist with substantially different marginal dependence on an earlier coordinate.

Again, these values are descriptive rather than causal definitions.

8.2 Quotienting endpoint labels

The four endpoint names carry no structural significance. Two labeled mappings are therefore placed in the same endpoint-label equivalence class when one can be obtained from the other by a permutation in $\text{Sym}(4)$ acting only on endpoint labels. Because every balanced map uses all four labels, this action is free, and the 2,520 labeled mappings yield

$$\frac{2520}{4!} = 105$$

unlabeled balanced partitions of the eight histories into four unordered pairs.

8.3 Time-preserving equivalence

For three binary coordinates, $G_{time} \cong (Z_2)^3$: each group element independently flips the labels $0 \leftrightarrow 1$ within any subset of the three fixed temporal coordinates. This group has size $2^3 = 8$. Acting on the 105 unlabeled endpoint partitions and identifying partitions in the same orbit leaves

35

fixed-time structural equivalence classes.

Thus even after endpoint labels and within-coordinate action names are discarded, identical two-bit endpoint capacity contains 35 different time-factored endpoint partition structures in this finite setting.

For comparison only, one may enlarge the acting group by also allowing all $3!$ permutations of the three coordinate positions. This deliberately discards temporal-coordinate identity and is therefore more permissive than the main interpretation. Under this enlarged action, the 105 partitions form

12

orbits. The residual multiplicity shows that the degeneracy is not exhausted by assigning an otherwise identical dependency to different coordinate names.

8.4 Coordinate-sensitivity summaries are still incomplete

For each intervention coordinate i , the three-dimensional binary cube contains four edges corresponding to changing only A_i .

Define

s_i = number of coordinate- i edges joining different endpoint classes.

Thus

$$s_i \in \{0, \dots, 4\}.$$

Across the 105 unlabeled balanced partitions there are only

17

distinct vectors

$$(s_0, s_1, s_2),$$

despite the 35 time-preserving structural classes.

Hence even supplementing endpoint capacity with the number of consequential one-coordinate interventions at every temporal coordinate does not reconstruct the complete partition organization.

The maximal case is particularly informative.

There are 16 balanced partitions with

$$(s_0, s_1, s_2) = (4, 4, 4).$$

In all 16, changing any single intervention coordinate in any context changes the endpoint.

Nevertheless, those 16 partitions occupy

10

different equivalence classes under time-preserving binary relabeling.

Therefore

Equal endpoint capacity plus equal maximal coordinate-wise sensitivity

still does not uniquely specify how histories are organized into endpoint equivalence classes.

This demonstrates that temporal causal structure is not exhausted by assigning a single influence magnitude to each time coordinate.

All numerical claims in this section were independently reproduced by direct exhaustive enumeration. The verification generates all balanced labeled maps, canonicalizes endpoint labels, constructs orbits under the stated

input actions, and computes coordinate-edge sensitivity directly; every count reported above was reproduced exactly.

9. General family

The ambiguity scales beyond the three-bit construction.

Consider

$$U = (A_0, \dots, A_{T-1}), A_i \in \{0, 1\}.$$

For any subset

$$S \subseteq \{0, \dots, T-1\}$$

with

$$|S| = k,$$

define

$$F_S(U) = U_S,$$

the projection onto the coordinates contained in S .

Every such system has

$$|\text{Im}(F_S)| = 2^k$$

and therefore

$$C_T(F_S) = k \text{ bits}.$$

Under uniform intervention,

$$H(E_T) = k \text{ bits}.$$

Yet coordinate i is endpoint-consequential exactly when

$$i \in S.$$

There are therefore

$$\binom{T}{k}$$

different coordinate-support patterns sharing the same deterministic endpoint capacity.

More generally, partitioning 2^T histories evenly among 2^k unlabeled endpoint classes gives

$$\frac{(2^T)!}{[(2^{T-k})!]^{2^k} (2^k)!}$$

balanced endpoint partitions.

Endpoint capacity remains only

k bits.

The combinatorial growth emphasizes the nature of the compression: a low-dimensional endpoint-control summary may correspond to a large family of differently organized intervention histories.

10. Relation to cognitive science and existing formalisms

The distinction established here should be interpreted relative to, rather than in competition with, richer cognitive, causal, temporal, and invariance frameworks. Its role is to separate an endpoint identification claim from stronger claims about temporal causal organization.

10.1 Cognitive causal learning, credit assignment, and empowerment

Credit assignment is a central cognitive and computational problem precisely because a later outcome does not automatically reveal which earlier action, state, or feature deserves causal credit. Human evidence shows that cognitive maps can filter outcome information according to causal relatedness during credit assignment (Moran et al., 2021), while temporal-credit-assignment work in reinforcement learning formalizes the problem as estimating the influence of earlier actions on later outcomes (Pignatelli et al., 2024). These literatures precedent the importance of temporally resolved causal relevance; the present result does not replace them.

Empowerment provides the complementary endpoint-control perspective. It treats causal influence as channel capacity from actions to future states or observations (Klyubin et al., 2005, 2008), and recent experiments show that children and adults use empowerment-related controllability and outcome variability when constructing causal models and choosing interventions (Yiu et al., 2026). Endpoint capacity and time-indexed intervention-effect support therefore answer different questions that can both matter in cognitive science: how much distinguishable future influence is available, and which earlier intervention contrasts remain consequential at the queried endpoint.

10.2 Generalized empowerment

Recent generalized empowerment explicitly extends the target from a final state to arbitrary future state sequences and permits independent variation of action duration, observation duration, and temporal separation (Abel et al., 2025). Accordingly, the claim that existing empowerment cannot represent temporal organization would be false. The present result instead explains why temporally resolved formulations contain information that an aggregate endpoint representation need not preserve.

10.3 Process empowerment

Process empowerment replaces fixed open-loop intervention sequences with feedback policies and is designed to characterize control robustly in stochastic environments (Tiomkin, Salge, & Polani, 2025). The stochastic construction in Section 7 does not compete with process empowerment. It shows only that permitting environmental noise does not eliminate the factorization issue when the representation under study remains invariant to arbitrary compound-input relabeling.

10.4 Causal information flow

Ay and Polani (2008) define information flow using interventions precisely to distinguish causal effect from correlational information measures. Such machinery is appropriate when a quantitative measure of the individual intervention contrasts defined above is required. The present paper therefore does not introduce a replacement measure for causal influence.

10.5 Directed and generalized directed information

Directed information provides a temporally ordered measure for interacting stochastic processes. More recently, generalized directed information has been introduced to permit source and target sequences occupying arbitrary temporal intervals; it is used to formalize both empowerment and its proposed mirror concept, plasticity (Abel et al., 2025). Temporal localization of information influence is therefore precedented. The present contribution is instead the endpoint-control/factorization application of the abstraction mismatch.

10.6 Structural input-output models

Computational mechanics and the ϵ -transducer characterize structured stochastic transformations between input and output processes by grouping histories according to their predictive consequences (Barnett & Crutchfield, 2015). Therefore history-equivalence partitions themselves are not new. The enumeration above uses such partitions only to expose how much structural variation may remain inside a fixed endpoint-capacity class.

10.7 Group invariance, orbits, and maximal invariants

The generic logic behind Proposition 1 belongs to established group-invariance theory. Under a group action, invariant functions are constant on orbits, while a maximal invariant is designed to distinguish those orbits as finely as possible. Modern treatments of group symmetry make the invariance/equivariance distinction explicit (Chiu, Sharp, & Bloem-Reddy, 2025). The paper does not claim this orbit logic as new mathematics. Its use here is specific: $G_{abstract}$ acts on compound intervention histories more coarsely than the factor-preserving action relevant to temporal-coordinate questions, and the P/M witness shows that the scientific support target can vary inside one abstract-input orbit.

10.8 Causal abstraction and exact transformations

Causal-abstraction work already establishes the broader principle that a coarse representation is adequate only relative to the intervention queries and causal distinctions it preserves. Rubenstein et al. (2017) formalize exact transformations between structural equation models by requiring consistency of intervention effects across levels of description. Beckers and Halpern (2019) develop increasingly restrictive notions of causal-model abstraction, including intervention mappings tied to state abstractions. These frameworks precede the general requirement that causal abstraction respect the interventions one intends to query.

Park et al. (2023) provide measure-theoretic causal-space foundations in which a causal space is built on a product measurable space and augmented by causal kernels that encode interventional structure. Buchholz, Park, and Schölkopf (2024) extend that framework by defining products of causal spaces and stochastic transformations between causal spaces, with direct causal interpretations as causally independent components, abstractions, and extensions. This directly precedes the broader point that variable/product structure and transformations among structured causal spaces can themselves carry causal meaning. The present paper does not claim ownership of that principle.

Xia and Bareinboim (2025) directly extend this broad precedent to lossy representations. Their projected-abstraction framework addresses cases in which multiple low-level interventions with different effects map to the same high-level intervention and analyzes how observational, interventional, and counterfactual causal queries can be translated under such loss. This makes explicit that preserving a coarse representation does not guarantee preservation of every intervention-sensitive target: causal adequacy is relative to the distinctions required by the query.

The present paper does not claim any of those general principles as new and does not supersede causal-space or projected causal abstraction. Its residual application is narrower and structurally different: every complete compound intervention history may remain individually present as a distinct atomic endpoint-channel input while the relational temporal product embedding among those otherwise distinguishable atomic histories is discarded. The lost information is therefore not necessarily atomic distinguishability itself; it can be the temporal factorization required to specify a fixed-coordinate intervention query. The resulting $G_{abstract}$ equivalence can

preserve endpoint-control properties while being too coarse for the fixed-coordinate support target defined relative to the narrower G_{time} action unless the product embedding is retained.

10.9 Variable relevance, essential variables, and edge sensitivity

In deterministic binary cases, Equation (8) is closely related to established variable-relevance or essential-variable ideas and to Boolean influence/sensitivity on the hypercube. Kahn, Kalai, and Linial (1988) are a canonical source for variable influence in Boolean functions. The present support relation is deliberately context-resolved: it records which controlled coordinate contrasts change the endpoint, whereas standard Boolean influence is typically an aggregate or probability-weighted summary over such contexts. Likewise, the finite counts S_i in Section 8 are cube-edge sensitivity counts, not a newly introduced causal-information primitive.

These connections narrow the ownership claim rather than weaken the endpoint-control result. Familiar factor-sensitive dependence objects are being used to expose what a factor-forgetting aggregate endpoint abstraction cannot identify.

11. What the result does and does not establish

The result establishes a constraint on inference from endpoint-level control descriptions:

Equal aggregate endpoint control need not imply equal time-indexed causal support.

It establishes this constructively in deterministic systems, preserves the distinction under stochastic degradation, and demonstrates substantial finite structural degeneracy under exact enumeration.

For a cognitive experiment or model, the result means that showing an intervention changes an endpoint can identify endpoint causal capacity without identifying the temporal factorization of that capacity. A claim about which earlier event remains causally relevant requires retaining the corresponding temporal coordinates or using a richer causal representation.

It does not establish that a cognitive system explicitly represents the support relation defined here.

It does not establish a psychological or neural credit-assignment mechanism, gate, selector, planning algorithm, temporal-depth estimator, or successor representation.

It does not establish that consciousness detects or controls the relevant distinction.

It does not establish that organisms or artificial agents prefer persistence over reconvergence.

It does not establish that behavioral possibilities are generated by the factorization-sensitive structure described here.

The formal result remains valid even if no cognitive system is behaviorally sensitive to temporal causal support; its immediate claim is methodological, not psychological.

12. Cognitive-science interpretation boundary

The formal separability motivates a bounded experimental question, not a claimed cognitive mechanism.

Suppose two task conditions are matched as tightly as experimentally possible on variables such as

reward ,

effort ,

immediate opportunity ,

future opportunity ,

and

aggregate endpoint control .

Suppose they differ only in whether an earlier causal contribution made by the participant or agent remains consequential for a later continuation or reconverges before that continuation.

One may then ask whether later choice, learning, or credit assignment differs between those conditions.

A representative downstream hypothesis is:

Temporal persistence or reconvergence of an agent's own causal contribution may influence later choice or learning even when reward, future opportunity, and endpoint control are matched.

This hypothesis is OPEN.

No empirical evidence for this hypothesis is supplied here.

The role of the formal result is narrower: it shows that such a manipulation is not logically equivalent to manipulating endpoint control. Temporal causal support can therefore be varied as an independent experimental factor rather than being silently absorbed into a generic quantity of future controllability.

A direct test would manipulate persistence versus reconvergence while independently controlling aggregate endpoint opportunity and control. An active-versus-yoked comparison could further test whether any effect depends on the agent's own causal contribution rather than merely on exposure to the corresponding event sequence.

Such experiments belong downstream of the present theoretical result.

13. Limitations

Several limitations are immediate.

First, the central theorem is an identification theorem about abstractions. It does not derive a new dynamical law or a cognitive mechanism. Its scientific value depends on whether distinguishing endpoint capacity from time-indexed support clarifies theoretical or empirical questions that would otherwise conflate them.

Second, the exact support relation and its time-preserving structural orbit are qualitative. In stochastic systems, two endpoint distributions may differ negligibly or substantially. Applications therefore require an established quantitative causal or statistical comparison appropriate to their intervention design.

Third, the finite enumeration considers small deterministic maps. Its purpose is exhaustive structural demonstration, not cognitive or biological realism.

Fourth, temporal support is not the same as complete causal provenance. A full dynamical causal model may distinguish mediation pathways, interaction terms, feedback dependence, and cross-boundary mechanisms that cannot be reconstructed from endpoint contrasts alone.

Fifth, time-indexed one-coordinate intervention sensitivity does not completely describe higher-order interaction structure. The enumeration itself demonstrates this: different endpoint partitions can share identical coordinate-sensitivity vectors. Synergistic causal contributions therefore require richer analysis than a vector of independent coordinate effects.

Finally, the result is conditional on a declared temporal factorization. If the temporal coordinates are scientifically meaningless and complete intervention histories genuinely should be treated as atomic symbols, then the additional structure identified here is irrelevant by construction. Conversely, when a cognitive theory assigns

distinct scientific roles to earlier versus later actions, cues, or perturbations, discarding that factorization can invalidate fixed-coordinate inferences.

That is not a failure of the theorem. It specifies when the distinction matters.

14. Falsification and novelty boundary

The mathematical application would fail if one could show that the claimed pair of systems is not actually equivalent under the aggregate endpoint-channel abstraction being invoked, or that the repaired factor-sensitive support target occupies the same G_{time} -orbit for the witness systems.

Neither is the case for the constructions above: P and M lie in the same abstract-input channel orbit, while their first-coordinate support is respectively full and empty, a structural distinction preserved under within-coordinate relabeling.

The generic mathematical principle used by Proposition 1 is not claimed as novel. Standard group-invariance/maximal-invariant theory already establishes that invariant representations are constant on group orbits and cannot recover targets that vary within those orbits. Mature causal-abstraction work already precedents the requirement that coarse causal descriptions preserve intervention distinctions relevant to the intended query. Park et al. (2023) and Buchholz et al. (2024) further precedent the causal significance of variable/product structure, while cognitive and reinforcement-learning literatures precedent the importance of causal relatedness and temporal credit assignment (Moran et al., 2021; Pignatelli et al., 2024). Yiu et al. (2026) additionally confirms that empowerment-related control cues are already being studied as part of human causal-model construction.

The ingredients are therefore strongly precedented:

- endpoint channel capacity as control is PRECEDENTED;
- temporal credit assignment as a cognitive and computational problem is PRECEDENTED;
- human use of causal/task structure in credit assignment is PRECEDENTED;
- intervention-based causal influence and temporally resolved information flow are PRECEDENTED;
- history equivalence, structural input-output models, and group-orbit/maximal-invariant logic are PRECEDENTED or established mathematics;
- causal adequacy of abstractions relative to intervention queries is PRECEDENTED;
- causal significance of variable/product structure, product-aware causal transformations, and deterministic variable influence are PRECEDENTED;
- empowerment as a control quantity and its use in human causal-learning research are PRECEDENTED;
- the existence of richer temporally resolved empowerment formulations is CONFIRMED.

The residual contribution carried forward here is deliberately narrower:

The residual contribution is limited to the specific endpoint-control mismatch between $G_{abstract}$ and G_{time} ; the exact P/M abstract-channel-isomorphism witness with full-versus-empty fixed-coordinate support; stochastic persistence under a shared post-channel; the exhaustive finite ambiguity analysis; and the general existence/scaling family.

Exact prior-art ownership of this integrated application remains OPEN. The manuscript therefore does not claim that the relationship is unprecedented, does not claim a new invariant-theory theorem, and does not claim a new information quantity or general theory of control. The cognitive-science conversion adds an application-level inference bridge, not a new mathematical result.

15. Conclusion

Cognitive science often asks what earlier parts of a process deserve causal credit for a later outcome. Endpoint control answers a related but different question: how much distinguishable causal influence a compound intervention history can exert on an endpoint.

When a sequence of temporally labeled interventions is collapsed into a compound channel input, endpoint capacity correctly measures that aggregate causal capacity. In doing so, however, it becomes invariant to arbitrary relabelings of complete intervention histories and therefore need not retain the temporal product structure required by a fixed-coordinate causal query.

The coordinate-labeled support sets transform under the narrower class of factor-preserving relabelings. Their structural orbit can therefore distinguish systems that are identical as abstract endpoint channels but differ in whether an earlier intervention remains consequential at the endpoint.

The result survives deterministic and stochastic dynamics and persists even after several aggregate quantities and coordinate-sensitivity summaries are matched. In cognitive terms, a matched endpoint or matched amount of control does not guarantee matched temporal causal credit.

Existing causal, credit-assignment, and temporally resolved information-theoretic formalisms can characterize the structure that endpoint capacity leaves unidentified. No new causal-information measure is required.

The bounded identification result can be summarized as:

Endpoint causal capacity does not identify its temporal factorization.

Accordingly, when a theory or experiment asks which earlier action, cue, or perturbation still matters at a later endpoint, the temporal factorization must be retained or independently recovered; endpoint equivalence alone is insufficient.

This constraint is compatible with established work on causal abstraction, credit assignment, empowerment, and temporal information flow. It specifies a boundary between what endpoint observations identify and what stronger process-level interpretations require.

Whether cognitive systems themselves represent or behaviorally exploit this distinction remains an empirical question.

Declarations

Data Availability Statement

Data sharing is not applicable to this article because no empirical datasets were generated or analyzed. The finite combinatorial enumeration is generated from the definitions stated in the manuscript and is reproducible from those definitions.

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